

Week 1

Project Name: Dungeon Quest RPG

1. Analysis Document

1.1 Project Overview

During Week 1, the project concept for Dungeon Quest was finalized as a Java-based dungeon exploration game where the player navigates through dungeon levels, fights enemies, collects items, and progresses toward escaping the dungeon.

1.2 Game Objectives

The following objectives were defined:

- ☐ Develop a structured dungeon exploration system
- ☐ Implement combat-based gameplay mechanics
- ☐ Design enemy behavior and interaction system
- ☐ Build inventory and item usage system
- ☐ Create modular OOP-based architecture using Java

1.3 Functional Requirements

The following core features were identified:

- ☐ Player movement system (grid-based or free movement)
- ☐ Enemy AI behavior system
- ☐ Combat system (attack, defense, health logic)
- ☐ Inventory system (items, weapons, healing)
- ☐ Level progression system

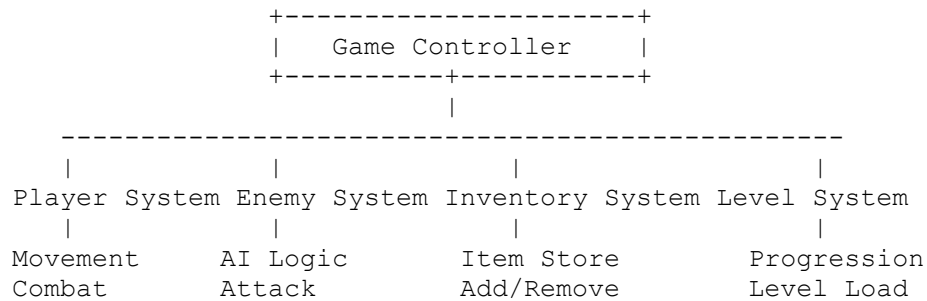
1.4 Non-Functional Requirements

- ☐ Code must follow Object-Oriented Programming principles
- ☐ System must be modular and scalable
- ☐ Game logic must be separated into classes
- ☐ Performance must support smooth gameplay loop

1.5 Technology Selection

- ☐ Programming Language: Java
- ☐ Architecture: Object-Oriented Design (OOP)

2. System Architecture Diagram (Week 1 Planning Level)



3. Initial Class Structure Blueprint (Conceptual)

Player

- health
- attack
- defense
- move()
- attackEnemy()

Enemy

- health
- damage
- AIBehavior()
- attackPlayer()

Item

- name
- effect
- use()

Inventory

- List<Item>
- addItem()
- removeItem()

LevelManager

- currentLevel
- loadLevel()
- nextLevel()

4. Game Concept Specification

Dungeon Quest was defined as a **level-based dungeon survival adventure game**. The player starts trapped inside a dungeon and must progress through multiple levels filled with enemies and obstacles.

Each level becomes progressively harder, requiring strategic combat and item usage. The final objective is to defeat the dungeon boss and escape.

Core gameplay loop

**Explore → Fight enemies → Collect items → Clear level →
Progress deeper**